College Football Bowl Game Predictor

UE24CS352A Machine Learning, Mini-Project Write-up
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1. Problem statement

Every year fans play bowl pick'em: choose the winner of each bowl game and give each pick a confidence from 1 to N, using each number once. A correct pick earns its confidence points. Winning needs both good picks and a good ordering of how sure you are. We build a model that, for a bowl game, predicts the score margin of team A over team B. The sign of the margin is the pick and its size is the confidence. This is the same task as the Stanford CS229 bowl predictor, which reports about 66% accuracy.

2. Dataset

Games. Regular-season results for 2003 to 2013 (8,365 games) from the sportsdataverse/cfbfastR-data repository. The log stops in mid December, so it contains no bowl games. **Bowls.** 356 bowl games (2003 to 2013, ties removed, each game counted once) from the bowlGames file in the geogunow/CollegeFootballBowlPredictions repository, matched to the game log through a team-name map plus four manual fixes. Every bowl team has at least 11 regular-season games.

A dataset we rejected. That repository also has teamrankings.com season stats (yards, turnovers, sacks, time of possession), which look like a perfect match for the Stanford features. They are not safe. For 375 of 477 bowl-team seasons, a team's published points per game is closer to its average *including* the bowl than to its regular-season average from the game log. As a sanity test, the rule "pick the team with the better record" scores 79% with the scraped records but 55% with records computed from regular-season games. A rule that simple should not be that good, so the scraped stats already contain the result. We dropped them (the check is in `leakage_check.py`). The cost is that we have no yardage or turnover features.

3. Approach

Features (7 per team, regular season only): win percentage, points scored and allowed per game, average margin with each game capped at 28 points, an opponent-adjusted rating, strength of schedule, and the margin over the last four games. The rating is a ridge regression of game margins on team indicators with a home-field term, solved separately for each season. **Model input** is the difference between the two teams' features. Every model is antisymmetric, so swapping A and B flips the prediction (linear models have no intercept; trees are averaged over both orderings; training data is mirrored).

Models. Two baselines (better record, better rating); linear regression on all features; ridge; linear regression with forward stepwise feature selection (3 features, chosen with an inner 5-fold CV), the same idea as the Stanford feature selection; random forest; gradient boosting; and logistic regression for the win probability. **Evaluation** is leave-one-season-out cross validation (train on ten seasons, test on the eleventh), as in the Stanford report. We report accuracy with a Wilson 95% interval, margin error, and the pick'em score when games in each season are ranked by predicted margin.

4. Implementation

Python with pandas, NumPy and scikit-learn. `build_dataset.py` creates the modelling table, `models.py` holds the antisymmetric model wrappers, `evaluate.py` runs the cross validation and writes tables and figures, and `predict.py` predicts one game with the season in question held out. `run_all.py` runs everything. Setup and commands are in the README.

5. Results and conclusions

Model	Accuracy (95% CI)	Margin MAE	Pick'em %
Baseline: better record	54.8% (50 to 60)		54%
Baseline: better rating	58.4% (53 to 63)		63%
Linear (all features)	59.6% (54 to 65)	12.8	65%
Ridge	60.4% (55 to 65)	12.8	65%
Linear + forward selection	59.3% (54 to 64)	12.7	65%
Random forest	53.9% (49 to 59)	13.5	57%
Gradient boosting	58.1% (53 to 63)	13.1	60%
Logistic regression	59.6% (54 to 65)		65%

Table 1. Leave-one-season-out results on 356 bowl games. Margin MAE is in points; predicting a margin of zero gives an MSE of 283, ridge gets 267. Pick'em % is points earned over the maximum for the same ranking (50% is chance for a random ranking).

Ridge was best at 60.4%, about 6 points above the better-record baseline and well above a coin flip ($p < 0.001$), but it is not clearly better than the better-rating baseline (58.4%); the intervals overlap heavily. Random forest and gradient boosting did not help, which is expected with only 356 games and 7 features. The confidence ranking is where the model earns its keep: splitting games into thirds by predicted margin, the least confident third is right 51% of the time, the middle 59% and the most confident 71%. (We chose ridge after seeing the results, so these numbers are a little optimistic.) Defense and schedule matter most: the largest ridge coefficients are points allowed, points scored and strength of schedule, but these features are correlated, so we do not read much into individual signs.

Conclusions. With only information available before the bowl, bowl games are hard to predict: we get about 60%, below the 66% reported by the Stanford project. We cannot tell whether their stats were pre-bowl, and ours is the stricter setup. The most useful finding is the leakage: season stats that include the bowl game turn a mediocre predictor into an apparent 80% one. Limitations are the small sample (an interval of about plus or minus 5 points), no yardage or turnover features, and no information on injuries, opt-outs or motivation. Future work is clean per-game box scores up to the end of the regular season, a team-level model of offense versus opposing defense, and an upset model for games the favorite is likely to lose.

